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RE: Siemens USA Response to NSF and OSTP's Request for Information on the Development of an Artificial Intelligence (AI) Action Plan.

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Introduction & Executive Summary

As a committed partner in American manufacturing, Siemens USA commends the administration for its commitment to developing an AI action plan prioritizing innovation, advancing competitiveness, and bolstering American technology leadership. Siemens is helping to pioneer America's industrial tech sector and is at the forefront of industrial artificial intelligence (Industrial AI). The company appreciates the United States' focus on AI as a national priority positioned to strengthen domestic manufacturing, empower the American workforce, and drive digitalization across all sectors.

Siemens is proud that the U.S. is the company's largest market, having invested more than \$90 billion in America over the past 20 years. Most recently, Siemens invested nearly \$700 million in new manufacturing facilities and factory expansions in North Carolina, South Carolina, California, Texas, Wisconsin, and New York. With 45,000 employees and 12,000 local suppliers in the United States, Siemens technology is deployed nationally, supporting critical infrastructure and vital industries forming the backbone of America's economy. With thousands of customers across all 50 states and Puerto Rico, Siemens is creating more productive factories, more resilient buildings, and more efficient power systems.

Harnessing expertise as the largest global provider of industrial automation hardware and software, Siemens deploys solutions with unique capabilities to combine the real and digital worlds. This domain know-how coupled with a proven track record of developing and advancing industry and infrastructure enables Siemens to provide valuable guidance to government leaders on the development of tech policy frameworks. The AI Action Plan will help to inform the federal focus on AI leadership, and Siemens sees Industrial AI applications as essential to this national priority.

Siemens plays a critical role in nearly every step of the AI value chain, from chip design to facility optimization to data centers to applications in every sector. The company is pleased to lend its technical expertise and perspective on Industrial AI to advance economic competitiveness, strengthen national security, and maximize the full potential of American manufacturing.

Summary of Key Recommendations:

- 1. Distinguish between Industrial AI and Consumer AI:** Federal leaders must recognize the unique role of Industrial AI in the AI ecosystem and acknowledge this distinction in any AI-related policy

action. Industrial AI applications differ significantly from consumer AI by relying on highly monitored, physics-based data to meet rigorous industrial standards. The advancement of these applications will empower the next generation of domestic manufacturing in the U.S., and it is critical to prioritize forward-thinking policies that will maximize Industrial AI's potential.

2. **Support the R&D and Innovation Ecosystem:** The strength of AI depends on the quality of computational resources and the ongoing research and development activities pushing American technology leadership to the cutting-edge. The federal government must dedicate funding to drive innovation through R&D programs, create a favorable environment to incentivize the private sector to further invest in R&D, strengthen support for industrial frontier models, and work across industry, academia and government to accelerate these investments.
3. **Invest in Hardware & Chips:** Semiconductors remain essential for AI development, and federal leaders have an opportunity to galvanize domestic onshoring for chip manufacturing by driving public and private sector investment in chip design and manufacturing. Digitalization and Industrial AI can enhance the semiconductor supply chain, offering end-to-end solutions for smart, secure, and reliable processes. The forthcoming AI Action Plan should prioritize policies that boost domestic chip capacity, promote digital twins, advance the semiconductor design ecosystem, support public-private initiatives, and integrate traceability principles.
4. **Strengthen Data Center Supply Chains:** Data centers are pivotal to the digital economy and AI development, driving technological innovation and economic growth. To maintain U.S. leadership in AI, the U.S. must prioritize the infrastructure and supply chains that power data centers. This includes ensuring the availability of essential electrical components and addressing industry challenges that result in long lead times and rising costs, two current obstacles to rapid data center expansion.
5. **Meet the AI Energy Challenge:** The energy demands of data centers are significant, driven by continuous operation and the computational power required for AI development. To meet these demands, fostering energy efficiency innovations and modernizing the U.S. energy system are essential. Policy solutions include energy efficiency R&D, grid infrastructure investments, accelerating grid interconnection, and integrating distributed energy resources, as well as convening industry stakeholders to develop regulatory frameworks and best practices that support new data center deployment.
6. **Advancing Digital Twins for National Security Impact:** AI-enabled digital twins are transforming national security by simulating and optimizing production in a digital environment before taking action in the real world. This technology leads to significant improvements in quality control and resilience in manufacturing, ensuring the U.S. can produce critical defense materials and technologies domestically. Siemens recommends utilizing digital twins to modernize national defense, improving transparency and accountability through Digital Product Passports, and integrating these solutions into DoD's overall modernization strategy.
7. **Bolstering the Workforce:** Generative AI, when applied in the industrial context, makes technologies more intuitive and allows workers to interact with machines using natural language, thereby empowering both experienced workers and new hires to find advanced career opportunities. To maximize Industrial AI's positive impact on the manufacturing workforce, the AI

Action Plan should include a focus on boosting skilled development programs and building meaningful partnerships across industry, academia, and government to address labor shortages and skill demands.

8. **Prioritizing Strong Digital Trade Protections:** The U.S. must emphasize strong digital trade rules that protect source code and algorithms, building upon the success of previous efforts to safeguard intellectual property and prevent forced data localization in trade agreements. These measures will support innovation by giving the U.S. technology sector the tools needed to compete globally and enable companies of all sizes to thrive in the new era of American manufacturing.

Detailed Response & Recommendations

Distinguishing between Industrial AI and Consumer AI

Siemens has been working on Industrial AI since the 1970s, when R&D teams first began incorporating AI into the company's products. Industrial AI differs significantly from AI used in commercial or consumer settings, as it must meet the rigorous requirements and standards of the most demanding industrial environments. With the ability to process large amounts of machine data and recognize complex patterns, Industrial AI helps organizations of all sizes accelerate their digital transformation at scale. Industrial AI stands to revolutionize manufacturing processes across product and production line design, operations, and a full suite of services.

Industrial AI is advancing automation and digitalization across industries, spanning power, automotive, aerospace, pharmaceutical, food & beverage, chemical, oil & gas, and more. With a digital-first approach built on massive volumes of data, Industrial AI is positioned to play an ever-increasing role in product design, testing, and creation to fully leverage the tremendous amount of industrial information generated in manufacturing processes. Thus, advancing the capabilities of Industrial AI is critical to securing domestic manufacturing, driving American economic growth, competing in the global economy, and advancing U.S. technological leadership.

To harness the full potential of Industrial AI, it is vital for the administration to distinguish the unique role of industrial applications from consumer-focused applications. The core distinction is that Industrial AI is trained on highly monitored data from sensors, machines, previous product designs, proprietary intellectual property, and other sensitive sources – providing a more reliable foundation for training AI models in this domain when compared to generally publicly available data used for many consumer-facing AI uses. Industrial AI is trusted, physics-based, and can be used to speak the complex languages of engineering and manufacturing. Furthermore, in industrial settings, manufacturers implement robust data quality checks and validation processes to ensure the accuracy and consistency of industrial data, which is critical for trustworthy AI.

As production becomes more connected and automated, Industrial AI is making human-machine collaboration more innovative, enabling manufacturing operators and engineers to find and analyze the most useful data points, which no human could ever fully analyze and interpret alone. This data analysis enables quicker decision-making that advances product customization and production efficiency. The administration has many opportunities to bolster and accelerate Industrial AI growth, including:

- **Boosting Understanding of Industrial AI:** Although AI has been deployed in some capacity in industrial settings for decades, there continues to be very little understanding of the impact and

unique uses of Industrial AI. Federal leaders should acknowledge the specific importance of Industrial AI in AI-related executive action, thereby deepening understanding throughout the policy ecosystem of the distinct operation of Industrial AI systems and the unique trustworthiness of the data they are trained on.

- Promulgating Narrower Definitions Related to AI: Especially as state legislatures across the country consider individual measures to regulate AI, targeted definitions matter more than ever. Overly broad definitions of which role an AI actor plays (developer, deployer, etc.) will negatively impact the ability of manufacturers to develop and provide access to next-generation technologies. In addition to being specific about the role of AI actors, it is essential to be very specific about risk. Most Industrial AI applications are utilized early in the value chain to optimize all stages leading up to finished products (design, testing, validation, etc.), and thus the applications are not necessarily deployed in the operational environment. Tailored definitions will be essential to avoid overburdening manufacturers as the data used in Industrial AI already meets higher standards since it is real-time, high-volume, and more trustworthy.
- Growing and Deepening Public-Private Partnerships: The federal government must focus on aligning public sector priorities with private industry efforts to maximize the impact of Industrial AI. The relationship between the public sector and the private sector is critical, and the entire AI ecosystem benefits from expanding understanding of fast-moving technology, finding creative ways to encourage broader adoption, and creating the necessary ecosystems to develop technology further.
 - Investing in Manufacturing USA Institutes: A strong example of a successful public-private ecosystem is Manufacturing USA. The network of Manufacturing USA Institutes is effective in improving the competitiveness of U.S. manufacturing by accelerating technology commercialization and developing the workforce. The administration should prioritize continued support for this program, especially for dedicated Institutes focused on AI in manufacturing.
- Aligning with international standards and voluntary AI frameworks: The U.S. should seek to assimilate new standards setting activities with internationally recognized standards and best practices that have already been crafted for AI risk management, including NIST's AI Risk Management Framework. Voluntary international technical standards are essential for fostering alignment, interoperability, and trust in AI systems both nationally and globally. By setting international benchmarks that are voluntarily adhered to by companies of all sizes, these standards provide structured processes for risk management and governance that facilitate innovation and technological advancement.

Industrial AI stands to empower and embolden the next generation of manufacturing in the United States. As the federal government remains committed to revitalizing this critical sector, the administration must prioritize forward-thinking policies to support Industrial AI and its ability to grow the economy, create jobs, empower innovation, and secure the next generation of American industry.

Supporting the R&D & Innovation Ecosystem

At Siemens, research teams have been dedicated to innovation and Industrial AI for decades, focusing on creating purposeful solutions to advance AI integration across core company technologies and competencies. Today, there are over 1,400 AI experts around the world and more than 500 active AI patent families at Siemens. As the federal leaders prioritize boosting AI innovation and fortifying U.S.

national R&D infrastructure, a dedicated focus on industrial applications and reliable use in manufacturing settings is critical. Siemens has focused R&D teams working to develop cutting-edge solutions for augmenting large language models (LLMs) to fit the industrial context, as well as to use existing technology to build security mechanisms inside the AI systems to maximize their reliability.

The growth of innovation in Industrial AI presents significant opportunity for transforming operational technology at the intersection of industry and AI. As operational environments are highly complex and rigid, Industrial AI – especially using generative AI – swiftly synthesizes knowledge of an operational system, combines it with planning information and engineering data, and allows operators to see their options for maximizing efficiency and optimizing productivity.

However, significant work remains to fully harness the potential of AI for industrial applications – starting at the model level. For domestic manufacturers to achieve meaningful gains in efficiency for industrial applications, investments must be made in AI models that are able to natively speak in the modalities required by industry. These models do not exist today. Manufacturers interact with data and information in modalities extending beyond what today's readily available LLMs are able to consume. Existing LLMs are trained to generate responses in text, images, videos, and computer source code. In contrast, the physics-based industrial frontier models that would be required to natively handle industrial data must understand and speak in the language of engineering and manufacturing – system design, mechanical design, electrical design, simulation, bills of materials, etc. The U.S. is well positioned to advance policies and seed an innovation ecosystem to optimize manufacturing for the benefit of every industry through greater efficiency. Siemens looks forward to building on industry partnerships to implement solutions and recommend the following:

- Supporting dedicated AI R&D programs focused on industrial applications: Across the federal ecosystem, agencies must ensure their R&D programs include specific focus on industrial applications for AI and frontier models. These resources are necessary over the long term to ensure models are created and continuously upgraded as industrial contexts are updated with new frontier developments in digitalization. By allocating resources towards industrial technologies and manufacturing ecosystems, federal agencies can create a favorable environment to incentivize the private sector to further invest in R&D. These efforts would be well positioned in several agencies including, but not limited to, the Departments of Commerce, Defense, and Energy, and the National Science Foundation.
- Convening public-private research collaborations: Especially when it comes to Industrial AI, a large share of the cutting-edge research and development is already taking place in the private sector. However, partnerships are crucial to driving new developments and capturing the full potential of Industrial AI. While there are several notable collaborations already underway between the private sector and academia, it would be constructive for the federal government to join Industrial AI-focused research partnerships to align priorities with the national policy agenda.
- Increasing access to federal data and computational resources: The ability to advance innovation for AI systems is dependent on access to large and diverse sets of both structured and unstructured data to effectively train models. A central challenge in building models – especially those needed to predictively maintain and predict improvements for a manufacturing environment – is gathering enough data to train robust AI systems. The administration should prioritize national efforts – in partnership with national labs and existing research institutes – to make more datasets available in machine-readable and industrial-deployable formats. These priorities must include support for shared, domain-specific testing environments and testing data to lower R&D costs and enable better validation.

- *Endorsing a shared national research resource for AI:* There is bipartisan congressional support for the National AI Research Resource (NAIRR), a shared national research infrastructure bringing together computational, data, software, AI models and training, and user support resources for research institutions and academia across the country. This shared resource will ensure companies of all sizes and researchers from all disciplines have access to advanced AI systems, computing power, testbeds, open-source models, and high-powered computational tools. It would be impactful for the administration to support the NAIRR or a similar alternative, as well to work with Congress to authorize this essential initiative to accelerate the pace of AI R&D in the U.S.
- *Providing broader resources for open-source data:* Open-source data and tools are essential for AI R&D, as they enable researchers and developers to build, test, and deploy AI models efficiently and at scale across applications. Several federal programs must continue to provide these valuable resources to the AI ecosystem, especially NIST's National Vulnerability Database, a critical resource for identifying and managing risks in software and hardware, including those related to AI systems.

U.S. AI leadership requires strong investment in national R&D infrastructure to explore new algorithms and models, continuously improve AI capabilities, and tailor uses across industries. The gains to be made in unleashing the full potential of Industrial AI and its applications are essential to fulfill the administration's commitment to onshoring domestic manufacturing. Siemens stands ready to partner with the federal government and across the AI ecosystem to maximize innovation and position the U.S. for economic growth and competitiveness.

Investing in Hardware & Chips

Semiconductors are critical for developing AI technology, especially as trends accelerate towards more advanced logic and memory chips. Federal leaders have an opportunity to drive continued investment for chips to be designed and manufactured in the U.S. For the domestic semiconductor manufacturing ecosystem to thrive, the federal government should prioritize the continued development of new manufacturing techniques and technologies through digitalization. Powered by Industrial AI, digital transformation applied to the semiconductor supply chain offers significant opportunity for end-to-end solutions for smart, secure, and reliable processes for designing and manufacturing next-generation chips.

The lifecycle of a semiconductor product is complex, and its complexity grows exponentially as semiconductors move through systems and across global supply chains. Digitalization offers a strong solution to supply chain concerns by introducing the opportunity for end-to-end traceability. Traceability provides the capability to supervise all transactions between data generators and to provide product provenance for chip-to-system design, development, and manufacturing. Manufacturers of all sizes can use traceability to track all inputs and manufacturing steps of all materials, thereby guaranteeing ultimate product security. As semiconductor products uses range from consumer devices to finished electronic products used in defense systems, this security is especially essential to verifying the sources of each component used in the end product.

As a global leader in semiconductor innovation and design, Siemens recommends the federal government continue to prioritize policies that boost domestic chip capacity, including:

- *Promoting digitalization and digital twins across semiconductor production:* The administration should encourage the digital transformation of the semiconductor industry. Some of the costliest areas of the semiconductor value chain lie in fabrication construction and semiconductor design – both of which can be best addressed by leveraging a comprehensive digital twin. A digital twin is a digital representation of a physical asset or process that evolves over the lifecycle, from a product or machine to production, plants or even the entire supply chain. Digital twins allow companies to manage the complexities of integrating software and hardware teams, integrate mechanical and production requirements, and achieve the necessary efficiency gains to be effective in the semiconductor ecosystem. The administration should support action that will expand the use of digital twins in addressing onshoring of the domestic semiconductor supply chain.
- *Nurturing the semiconductor design ecosystem:* In addition to incentives for the creation and proliferation of semiconductor manufacturing in the U.S., the administration must focus investment and support on semiconductor design. While less capital intensive than manufacturing, semiconductor design represents a key point of enablement for the foundational innovations that underly AI. Private sector focus on designs optimized for AI workloads and applications is growing consistently. Timely and aggressive emphasis from the federal government on semiconductor design solutions developed in the U.S. will ensure a healthy innovation pipeline for the fast-changing future.
- *Continuing support for public-private initiatives on semiconductor manufacturing:* As previously mentioned, the Manufacturing USA Institutes program has been very successful in increasing domestic manufacturing competitiveness and promoting a robust R&D infrastructure. The administration must continue to support this program in the semiconductor domain, especially to launch dedicated initiatives for semiconductor manufacturing. These SMART (Semiconductor Manufacturing and Advanced Research with Twins) USA Institutes will significantly enhance the domestic semiconductor industry by deploying digital twins to improve design, manufacturing, advanced packaging, assembly, and test processes.
- *Integrating trusted traceability principles in the semiconductor supply chain:* Siemens appreciates the administration's strong focus on bolstering the U.S. domestic supply chain. The administration should support the allocation of federal resources towards specific activities to secure and enhance the supply chain – especially traceability. Federal resources and funding must be designated towards creating standards for trusted traceability and program implementation, as well as incentives for companies to actively participate in the implementation of trusted supply chain traceability infrastructure.

Strengthening Data Center Supply Chains

In today's digital economy, data centers are core drivers of growth and technological innovation. They house the computer systems storing and managing the vast amounts of data generated daily – data essential for the functioning of commerce, online services, social media, and more.¹ Data centers are also integral to AI development, providing the computational requirements, real-time processing capabilities, and cloud services needed for machine and deep learning. As the digital economy continues to grow and AI becomes increasingly integrated into daily lives, global demand for data

¹ The U.S. data center industry's direct contribution to U.S. Gross Domestic Product (GDP) increased 53 percent between 2017-2021, from \$89 billion to \$136 billion, according to a study by PWC. *Economic, Environmental, and Social Impacts of Data Centers in the United States*, PWC (Sept. 2023), <https://www.centerofyourdigitalworld.org/2023-impact-study>.

center capacity could rise at an annual rate of 19 to 22 percent between 2023 and 2030, reaching an annual demand of 171 to 219 gigawatts.²

To support continued buildout of data centers here in the United States—and continued global leadership in AI—the federal government must prioritize the infrastructure and supply chains needed to power data centers. The energy needs of large-scale data centers are immense, requiring reliable and uninterruptible power supply. To deliver that energy, data centers require a robust ecosystem of electrical distribution equipment and grid technologies, including circuit breakers, panel boards, switchgear, switchboards, substations, monitoring devices, transformers, and software to ensure data center safety, efficiency, and reliability. The availability of electrical components is therefore a key concern for data center developers. Long lead times and rising costs for these technologies are a potential obstacle to the rapid growth of data center construction in the U.S.

The U.S. electrical manufacturing industry has moved boldly to support the acceleration of data center construction and AI development. Siemens alone has invested nearly \$700 million in recent years to build high-tech manufacturing facilities in the U.S., including medium- and low-voltage power equipment used in data centers and other critical infrastructure. In March 2025, Siemens cut the ribbon for the company's newest facility, investing \$190 million in Fort Worth, Texas to produce low-voltage switchboards for data center customers, an investment to include a total of 800 new jobs to the local economy by next year.³ Additionally, Siemens recently celebrated the opening of its new Pomona, California manufacturing facility adjacent to its legacy site, increasing the capacity on campus to manufacture vital low-voltage electrical equipment, such as switchboards and lighting panels.⁴ This site enables Siemens to better meet demand from industrial, commercial, and construction markets, with outsized need from America's booming data center industry. Siemens also has plants in Grand Prairie, TX, Broken Arrow, OK, Spartanburg, SC, and Wendell, NC, each manufacturing electrical technologies for data centers.

Siemens and other electrical manufacturing industry leaders are eager to continue investing in U.S. supply chains and work with the federal government to power the future of the U.S. AI industry. Due to the importance of electrical components to the data center and AI sectors, Siemens recommends the following:

- *Industry Convenings to Support Data Center Supply Chain Security*. The White House, in coordination with the Department of Energy, Department of Commerce, and other agencies, could host industry-led convenings on critical electrical and grid components (including software, transformers, switchgear, circuit breakers, switchboards, and other technologies) to examine policy, regulatory, and market challenges facing electrical manufacturing supply chains for data centers. These convenings should include manufacturers, data center developers, electric utilities, and other private-sector stakeholders, and should seek to identify policy recommendations to support the expansion of electrical manufacturing capacity both in the U.S. and among allies to meet the infrastructure needs of AI and data center growth.

² Bhargh Srivathsan, et al., *AI power: Expanding data center capacity to meet growing demand*, McKinsey & Company (Oct. 2024), <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/ai-power-expanding-data-center-capacity-to-meet-growing-demand>.

³ *Siemens officially opens doors on \$190M Fort Worth manufacturing hub to support AI infrastructure boom*, Siemens (March 2025), <https://www.siemens.com/us/en/company/press/press-releases/smart-infrastructure/siemens-opens-190m-ft-worth-mfg-hub-ai-infrastructure.html>.

⁴ *Siemens Invests \$95 Million to Expand and Upgrade Legacy Pomona, CA Facility, Increases Manufacturing Capacity for Critical U.S. Energy Infrastructure*, Siemens (March 2025), <https://www.siemens.com/us/en/company/press/press-releases/smart-infrastructure/siemens-invests-95m-expand-pomona-facility.html>.

Through collaboration between the federal government and industry, the U.S. can ensure AI infrastructure is built at the speed and scale needed to win the AI race.

Meeting the AI Energy Challenge

One of the biggest challenges facing the U.S. in its efforts to accelerate innovation in AI is the energy needs of data centers. Data centers require significant power because they house thousands of continuously operating servers and networking equipment which process and store massive amounts of data. The scale of data processing and the need for uninterrupted operation—as well as the robust cooling systems required to prevent servers and other assets from overheating—drive high energy consumption. Moreover, developing next-gen AI models, including LLMs, is an energy-intensive process due to the computational power needed to train them on vast datasets across multiple graphics processing units (GPUs).

As AI adoption and innovation continues to scale, data center power needs are likely to increase substantially. One study estimates data center demand is expected to grow to about three times higher than current capacity by the end of the decade, rising from 3 to 4 percent of total U.S. power demand today to between 11 and 12 percent in 2030.⁵ Data center energy needs, coupled with surging demand from manufacturing and electrification countrywide, are driving the U.S. into a new era of power sector load growth.

Energy availability is already a major factor in determining whether—and how quickly—new data centers can come online. The time required to establish new power connections for data center sites in major hubs around the country is increasing, with lead times in critical regions reaching three years or longer.⁶

To meet the energy challenge, the U.S. must foster new innovations in efficient energy infrastructure, as well as leverage grid and energy technologies to quickly power new data center builds.

A. Fostering Energy Efficiency Innovation

Energy efficiency within the data center ecosystem can promote a range of benefits, including optimizing spend and resource savings, improving data center reliability, and minimizing impacts to the grid. Tech companies and data center operators invest heavily in energy efficiency innovation to capture these benefits. Data center efficiency practices and technologies include cooling and thermal optimization; energy/demand monitoring and management software; heat recovery systems to capture and reuse waste heat generated by data center equipment; energy-efficient servers, storage devices, and other hardware; and virtualization to reduce the number of physical services required.

Siemens is an industry leader in supporting data center energy efficiency. For example, Siemens offers energy management software, providing data center operators with real-time analytics and diagnostic maintenance insights. Siemens' building management platforms, acting on data from a single interface, promote visibility and further efficiency gains. Siemens' solutions include software to interface with the interconnecting utility for demand response and reliability support—enabling a synergistic relationship between data centers and associated utilities. Machine learning also plays

⁵ Alastair Green, et al., *How data centers and the energy sector can satiate AI's hunger for power*, McKinsey & Company (Sept. 2024), <https://www.mckinsey.com/industries/private-capital/our-insights/how-data-centers-and-the-energy-sector-can-satiate-ais-hunger-for-power>.

⁶ *Id.*

an important role in optimizing energy use by better managing temperature-regulating devices in air and water systems.

Additionally, Siemens' White Space Cooling Optimization uses AI to apply the right amount of cooling when and where it is needed among data center racks. For this solution, Siemens deploys a network of sensors, cooling unit controls, and an AI engine to dynamically match facility cooling to data centers' real-time IT load. White Space Cooling Optimization addresses the challenges of managing processes with minimal on-site staff, while optimizing performance and increasing reliability. It is a cost-effective solution that can be fully deployed in just a few weeks, in almost any data center, with minimal on-site intrusion.

Siemens also enables energy efficiency in the planning and design phase of data centers by using digital twin software to create, maintain, and visualize digital twins of buildings and infrastructure. Digital twins can simulate various data center cooling strategies, optimize server utilization to minimize energy waste, and perform predictive maintenance in data center infrastructure to avoid potential issues before they lead to system inefficiencies or failures.

As AI continues to evolve and data centers expand to meet growing computational demands, new innovations in energy efficiency technologies will be needed. Siemens stands ready to work with experts at the Department of Energy and across the federal government to advance data center efficiency. Siemens recommends the following policy options:

- *Energy Efficiency RD&D*: The federal government should invest in research, development, and deployment (RD&D) initiatives within the Department of Energy and the National Laboratories to support new breakthroughs in data center energy efficiency. Potential research areas should include advanced cooling technologies, energy management software, demand response software, and AI algorithm optimization. Moreover, the Department of Energy should work to ensure research uptake and commercialization by developing pilot programs to support next-gen energy efficiency technologies.
- *Private Sector Initiatives to Enable Energy Efficiency Best Practices*: Department of Energy-led initiatives aimed at facilitating private sector collaboration on energy efficiency, including the Better Buildings Initiative, are important for sharing best practices and enabling partnership opportunities. The federal government should continue to support these initiatives and ensure they are incorporating the latest innovations in energy efficiency technologies. To encourage new innovation, the Department of Energy could consider creating a contest to explore new energy efficiency technologies and methods for reduction of power, utilization of waste heat, etc.

Ultimately, private and public sector investment in energy efficiency will enable a more resilient and scalable foundation for future data center growth.

B. Unlocking the U.S. Energy System's Full Potential

Over the last two decades, electricity demand growth in the U.S. has mostly been stagnant. Today, analysts are projecting that nationwide demand could rise 128 GW over the next five years, an increase of 8.2 percent.⁷ Most of this demand in the short-term is coming from new data center builds and manufacturing.

⁷ John D. Wilson, et al., *Strategic Industries Surging: Driving US Power Demand*, Grid Strategies (Dec. 2024), <https://gridstrategiesllc.com/wp-content/uploads/National-Load-Growth-Report-2024.pdf>.

A central issue in the AI race is whether the U.S. energy system will be able to generate enough electricity to power—and effectively manage—new data center loads. There are many potential roadblocks. Much of the existing transmission and distribution infrastructure in the U.S. lacks capacity to manage increasing load. New energy generation is difficult to permit and to come online, with over two terawatts of new generation and storage projects waiting in interconnection queues.⁸ Moreover, significant financial investment is required to build new power plants, upgrade existing infrastructure, and integrate renewable energy sources. Valid challenges around ratepayer equity and how to finance this multi-terawatt infrastructure are further clouding the path to infrastructure buildout and risking the time-sensitive realization of additional power, energy, and transmission infrastructure on the grid.

As noted above, access to power is emerging as a critical bottleneck for new data center builds, and data center operators are contending with grid constraints. This results from a mismatch between the development period for a typical greenfield data center and the time required to complete associated interconnection studies and infrastructure upgrades, which can span as long as four to eight years.⁹ Researchers found lead times for interconnection are the foremost consideration for data center operators when building new sites.¹⁰

The federal government can play a critical role in addressing energy system challenges, including by exploring the following policy options:

- Convening Energy and Data Center Stakeholders: The Department of Energy should convene data center developers and operators, energy utilities, technology companies, and other key stakeholders to develop a roadmap to address energy supply bottlenecks and develop strategies to promote U.S. AI leadership. These convenings should address critical permitting and regulatory framework challenges—and should develop solutions to accelerate the pace of AI and data center infrastructure buildout.
- Grid Modernization: The federal government should facilitate investments in grid infrastructure modernization and smart grid technologies and assess opportunities for reconductoring and integration of grid-enhancing technologies, including dynamic line rating, to increase the power carrying capacity of existing transmission infrastructure.
- Enabling DER Solutions: Microgrids, onsite battery storage, and other distributed energy solutions (DERs) can support data center resilience and enable data centers to provide critical grid services. The Department of Energy should evaluate these technologies and offer technical assistance to grid stakeholders to support novel backup power strategies and leverage its technical expertise to work with the Federal Energy Regulatory Commission, states, and grid stakeholders to formulate frameworks and best practices to support such innovation. The Department of Energy should examine how utilities and data center operators can synergistically control energy supply to maximize resilience through software technologies for demand management and microgrid ramping and control strategies.
- Facilitating Co-location of Energy Infrastructure: Facing grid constraints, data center developers have moved quickly to identify creative ways to develop on-site generation and grid

⁸ Joseph Rand, et al., *Queued Up: 2024 Edition*, Lawrence Berkeley National Laboratory (April 2024), https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_1.pdf.

⁹ Vivian Lee, et al., *Breaking Barriers to Data Center Growth*, BCG (Jan. 2025), <https://web-assets-pdf.bcg.com/prod/breaking-barriers-data-center-growth.pdf>.

¹⁰ Green, *supra* note 5.

infrastructure. The Department of Energy should consult with states, utilities, data center developers, and other stakeholders to identify the regulatory, financial, and technical constraints for effective co-location of energy assets. As with the recommendation above, the Department of Energy should seek to develop frameworks and best practices to support innovative co-location methods.

- *Accelerating Interconnection*: The Department of Energy and National Laboratories should promote RD&D, pilot programs, and technical assistance initiatives to foster AI and software solutions aimed at streamlining grid interconnection processes and associated planning studies.
- *Supporting Energy System Planning*: The Department of Energy and the National Laboratories should work with technology companies to develop and promote planning tools for projecting future infrastructure needs to support data centers and other large loads (recognizing both potential load growth and flexibility) and to expand technical support to state energy and utility planning departments.

Notably, Siemens is well-positioned to address some of the critical energy system challenges outlined above. Siemens' comprehensive software portfolio includes advanced grid software products, IoT-based applications, grid automation and protection solutions, as well as teams of software engineers, solutions architects, AI/ML experts, and project managers all with utility industry expertise. Siemens software helps utilities leverage real-time grid data, cloud-based simulations, and AI to better plan, operate, and manage their grids—thus enabling utilities to optimize their investment and get the most out of their existing grid infrastructure. Moreover, Siemens helps deploy and manage distributed energy systems, including microgrids, which can be leveraged by data center operators to support energy resilience and provide critical grid services. Lastly, Siemens Financial Services – the company's investment arm – is driving private sector capital to critical energy generation, battery storage, and carbon capture projects.

Looking ahead, Siemens hopes to work closely with the federal government, states, and other key stakeholders to meet energy demand growth and secure U.S. AI leadership.

Advancing AI-Enabled Digital Twins for National Security Impact

The integration of artificial intelligence in manufacturing is not only transforming industrial capabilities but also strengthening national security. Advanced AI technologies enhance efficiency, improve quality control, and optimize supply chain operations, making the manufacturing sector more resilient and competitive. This resilience ensures the U.S. can produce critical defense materials and technologies domestically.

Digital twins leverage highly monitored data from sensors, machines, and Industrial AI to design, simulate, and optimize production in the digital world before taking action in the real world. This cutting-edge technology helps manufacturers of all sizes and disciplines to tackle industry's biggest challenges, and it's happening right here in the U.S.

Digitalization adoption allows for a comprehensive suite of products and operations to be fully integrated in a digital backbone of operational data via the Industrial Metaverse. The Industrial Metaverse is a world powered by Industrial AI, where real machines and factories, buildings and cities, grids and transportation systems are mirrored in the virtual world. In a digital environment, problems can be identified, analyzed, and fixed quickly – or better yet, discovered before they arise. Siemens

brings this capability to the national security environment and is ushering in the next generation of technologies that should be utilized for national security.

Additionally, through digitalization, a Digital Product Passport is possible and functions as a digital record for products, documenting their entire lifecycle, from the initial sourcing of raw materials throughout production and recycling. It contains information about the product's composition, manufacturing, performance, and resources required for production. The information can be accessed for example, through a QR-code and can help to increase transparency and accountability in an array of industries.

- *Modernizing National Defense with Digital Twins*: To fully modernize the federal government, the Department of Defense, DoD program offices, and other relevant agencies should utilize industry-proven digital twin solutions to enhance national security and optimization of the defense community. More broadly, digitalization should be incorporated into the overall modernization strategy of the federal government to improve efficiency and streamline productivity across the federal ecosystem.

Bolstering the Workforce

Major technology developments raise questions for workers, especially in the realm of AI applications. As the federal government considers how to ensure AI enhances the American economy, Industrial AI stands as a testament to the potential for technology to galvanize the workforce instead of replacing it. Industrial AI expands what is humanly possible, creating more opportunities for workers of all skill levels and providing growth opportunities at all stages of their careers.

When applied in manufacturing settings, Generative Industrial AI (Gen AI) is making technologies more intuitive by allowing workers to 'speak' to machines using natural language instead of code. Siemens has led the industry in this area, deploying the first Gen AI-powered assistant, or 'copilot', for industrial operations. For experienced workers, industrial copilots help them with more repetitive tasks, so they can spend more time innovating and making decisions. For new hires, industrial copilots help people start manufacturing jobs without specialized skills. This can bring new people into the workforce, helping fill more of America's 500,000 open positions in manufacturing.

To maximize Industrial AI's positive impact for the American workforce, the administration can prioritize several areas, including:

- *Investing in Workforce Development*: To close gaps in the domestic manufacturing workforce, the administration must invest in the workforce development programs necessary for today and the future. From high school STEM classes to university laboratories, institutions are developing curricula to tackle real-world challenges through programs that teach both technical and career skills. This collective action is exactly what it takes to develop the digital talent required for continued U.S. leadership in technology and industry. Workforce development programs – ranging from apprenticeships to WIOA initiatives to on-the-job training – must incorporate a strong focus on advanced manufacturing jobs, leveraging Gen AI to boost a stronger skilled workforce for the domestic manufacturing sector.
- *Building Meaningful Partnerships Across the Ecosystem*: As is the case with many solutions, partnership is the new leadership. The strongest workforce successes will occur when industry, academia, and the government work together to support skills cultivation from the student level throughout the rest of their careers. There are numerous collaborative efforts across the country

working to address critical labor shortages and rapidly shifting skill demands. As an example, Siemens recently kicked off a microcredentialing program with the University of Colorado Boulder and Pennsylvania State University to credential students and recognize career readiness, thereby creating a stronger pipeline of engineering talent demanded by industry. These credentials are intended to supplement a formal degree by validating industry knowledge and skills. The administration must prioritize support for these cross-sector partnerships to propel innovative solutions for closing the skills gap and driving job creation for the future.

Prioritizing Strong Digital Trade Protections

Federal leaders can further advance Industrial AI by prioritizing strong rules for digital trade, especially by upholding and advancing strong protections for source code and algorithms. Policymakers should build upon the success of U.S.-led efforts to protect intellectual property, prevent forced data localization, and oppose the use of customs duties on electronic transfers within sectoral trade agreements and full Free Trade Agreements. Such support for innovation and U.S. technology from the U.S. government gives industry the confidence and protections to serve global markets. These priorities have built the foundation of an ecosystem where American-driven technology can compete, grow, and innovate in the global economy. By remaining committed to strong digital trade protections, the U.S. can propel Industrial AI applications to enable all companies – including startups, small- and medium-sized enterprises, and industrial giants – to thrive in this new era for American manufacturing.

Conclusion

Siemens is committed to advancing U.S. manufacturing and pioneering the future of industrial technology. Industrial AI is essential to digitalizing the manufacturing sector, growing domestic capacity, and achieving the administration's core priorities of economic dominance and national security. As federal leaders seek to solidify and enhance American technology leadership, Siemens looks forward to continued opportunities for guiding these efforts to fortify the innovation ecosystem, strengthen data center supply chains, and harness the full potential of Industrial AI to revolutionize the next generation of American manufacturing.

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